

Momentum Strategies

Momentum strategies capitalize on the tendency of assets that have performed well (or poorly) to continue that performance in the near future. These strategies involve identifying assets with strong directional movement and riding that trend until signals indicate a potential reversal.

During the 2020-2021 bull market in technology stocks, momentum strategies performed exceptionally well. Companies like Tesla (TSLA), which rose over 700% in 2020, created ideal conditions for momentum algorithms to capture significant gains by identifying and following the powerful uptrend.

A simple momentum check in Python:

```
1. # Simple momentum check
2. last_month_price = 100
3. current_price = 120
4.
5. # Calculate the percentage change
6. percent_change = (current_price - last_month_price) / last_month_price * 100
7.
8. # If price increased by more than 10%, it has positive momentum
9. if percent_change > 10:
10.     print("Strong positive momentum - potential buy signal")
11. elif percent_change < -10:
12.     print("Strong negative momentum - potential sell signal")
13. else:
14.     print("No strong momentum detected")
15.
```

Machine Learning-Based Strategies

The rise of machine learning has transformed algorithmic trading, allowing systems to identify complex patterns that traditional statistical methods might miss. Python's dominant position in machine learning makes it the natural choice for these advanced strategies.